Physica bk5

May 3, 2026

1 On GlossySnake

A LaTeX Proof-of-Concept of the GlossySnake project. By the nature of the machine-glossing, the inaccurate and malformatted glossing is expected.

2 Physica bk5

PHYSICA BK.V

PHYSICS BOOK V

Κεφάλαιο 1

Chapter 1

Μεταβάλλει δὲ τὸ μεταβάλλον πᾶν τὸ μὲν κατὰ συμβεβηκός,

It changes but the changing all the on one hand according to accident,

οἷον ὅταν λέγωμεν τὸ μουσικὸν βαδίζειν, ὅτι ᾧ συμβέβηκεν

such as when we say the musical walking, because to which it has happened

μουσικῷ εἶναι, τοῦτο βαδίζει· τὸ δὲ τῷ τούτου τι

according to music, to be, this is walking. The but to the this something

μεταβάλλειν ἁπλῶς λέγεται μεταβάλλειν, οἷον ὅσα λέγεται

to change simply is said to change, such as whatever is said

κατὰ μέρη (ὑγιάζεται γὰρ τὸ σῶμα, ὅτι ὁ ὀφθαλμὸς ἢ ὁ

according to parts (is healthy for the body, because the eye or the

θώραξ, ταῦτα δὲ μέρη τοῦ ὅλου σώματος)· ἔστι δέ τι ὃ

thorax, these but parts of the whole body). There is but something that

οὔτε κατὰ συμβεβηκὸς κινεῖται οὔτε τῷ ἄλλο τι τῶν
neither according to accident moves nor by the other something of the
αὐτοῦ, ἀλλὰ τῷ αὐτὸ κινεῖσθαι πρῶτον. καὶ τοῦτ' ἔστι τὸ καθ'
same, but by the itself to move first. And this is the by
αὐτὸ κινητόν, κατ' ἄλλην δὲ κίνησιν ἕτερον, οἷον ἀλλοιωτόν,
itself movable, according to another but motion another, such as a changeable,
καὶ ἀλλοιώσεως ὑγιαντὸν ἢ θερμαντὸν ἕτερον. ἔστι δὲ καὶ
and of alteration health-giving or heat-giving another. there is but also
ἐπὶ τοῦ κινοῦντος ὡσαύτως· τὸ μὲν γὰρ κατὰ
upon the mover likewise: the on the one hand for in accordance with
συμβεβηκὸς κινεῖ, τὸ δὲ κατὰ μέρος τῷ τῶν
the contingent moves, the on the other hand according to part to the of those
τούτου τι, τὸ δὲ καθ' αὐτὸ πρῶτον, οἷον ὁ μὲν ἰατρὸς
things something, the but in regard to itself first, such as the one doctor
ἰᾶται, ἡ δὲ χεὶρ πλήττει. ἐπεὶ δ' ἔστι μέν τι τὸ κινοῦν
heals, the and hand strikes. since there is it is indeed something the moving
πρῶτον, ἔστι δέ τι τὸ κινούμενον, ἔτι ἐν ᾧ, ὁ χρόνος,
first, there is but something the moved, still in which, the time,
καὶ παρὰ ταῦτα ἐξ οὗ καὶ εἰς [224b] ὅ—πᾶσα γὰρ κίνησις
and besides these out of which and into [224b] all-every for motion
ἔκ τινος καὶ εἴς τι· ἕτερον γὰρ τὸ πρῶτον κινούμενον
comes from something and to something. another for the first moved
καὶ εἰς ὃ κινεῖται καὶ ἐξ οὗ, οἷον τὸ ξύλον καὶ τὸ
and into that which is moved and out of which, such as the wood and the
θερμὸν καὶ τὸ ψυχρόν· τούτων δὲ τὸ μὲν ὅ, τὸ δ' εἰς
hot and the cold. Of these but the on one hand that, the but into
ὅ, τὸ δ' ἐξ οὗ—ἡ δὴ κίνησις δῆλον ὅτι ἐν τῷ ξύλῳ, οὐκ
that, the but out of which-the indeed movement is clear that in the wood, not

ἐν τῷ εἴδει· οὔτε γὰρ κινεῖ οὔτε κινεῖται τὸ εἶδος ἢ ὁ τόπος

in the form. Neither for moves nor is moved the form or the place

ἢ τὸ τοσόνδε, ἀλλ' ἔστι κινοῦν καὶ κινούμενον καὶ εἰς ὃ

or the such but it is moving and being moved and into that which

κινεῖται. μᾶλλον γὰρ εἰς ὃ ἢ ἐξ οὗ κινεῖται ὀνομάζεται

is moved. more for into that which or out of which is moved is called

ἡ μεταβολή. διὸ καὶ ἡ φθορὰ εἰς τὸ μὴ ὂν μεταβολή

the change. Therefore and the decay into the not being transformation

ἐστιν· καίτοι καὶ ἐξ ὄντος μεταβάλλει τὸ φθειρόμενον· καὶ ἡ

is; and yet also from being transforms the corruptible; and the

γένεσις εἰς ὄν, καίτοι καὶ ἐκ μὴ ὄντος. τί μὲν οὖν ἐστιν

generation into being, and yet also from not being. what indeed therefore is

ἡ κίνησις, εἴρηται πρότερον· τὰ δὲ εἴδη καὶ τὰ πάθη καὶ ὁ

the motion, has been said before; the but types and the qualities and the

τόπος, εἰς ἃ κινοῦνται τὰ κινούμενα, ἀκίνητά ἐστιν, οἷον ἡ

place, into which are moved the moved, unmoved is, such as the

ἐπιστήμη καὶ ἡ θερμότης. καίτοι ἀπορήσειεν ἄν τις,

knowledge and the heat. And yet it would puzzle someone, a certain one,

εἰ τὰ πάθη κινήσεις, ἡ δὲ λευκότης πάθος· ἔσται γὰρ

if the qualities are movements, the but whiteness is a quality; it will be for

εἰς κίνησιν μεταβολή. ἀλλ' ἴσως οὐχ ἡ λευκότης κίνησις,

into motion transformation. But perhaps not the whiteness is a motion,

ἀλλ' ἡ λεύκαν σις. ἔστιν δὲ καὶ ἐν ἐκείνοις καὶ τὸ κατὰ

but the whiteness is. it is moreover and in those and the according to

συμβεβηκὸς καὶ τὸ κατὰ μέρος καὶ [τὸ] κατ' ἄλλο καὶ

the incident and the according to part and [the] according to another and

τὸ πρώτως καὶ μὴ κατ' ἄλλο, οἷον τὸ λευκαινόμενον εἰς

the first and not according to another, such as the becoming white into

μὲν τὸ νοούμενον μεταβάλλει κατὰ συμβεβηκός (τῷ
on the one hand the conceived changes according to the incident (the
γὰρ χρώματι συμβέβηκε νοεῖσθαι), εἰς δὲ χρῶμα ὅτι μέρος
for color has occurred to be understood), into but color that part
τὸ λευκὸν τοῦ χρώμα τος (καὶ εἰς τὴν Εὐρώπην ὅτι μέρος αἱ
the white of the color the (and into the Europe that part the
Ἀθῆναι τῆς Εὐρώπης), εἰς δὲ τὸ λευκὸν χρῶμα καθ' αὐτό.
Athenians of Europe), into but the white color according to itself.
πῶς μὲν οὖν καθ' αὐτὸ κινεῖται καὶ πῶς κατὰ
how on the one hand therefore according to itself moves and how according to
συμβεβηκός, καὶ πῶς κατ' ἄλλο τι καὶ πῶς τῷ αὐτὸ
the incident, and how according to another something and how to the it
πρῶτον, καὶ ἐπὶ κινοῦντος καὶ ἐπὶ κινουμένου, δῆλον, καὶ ὅτι ἡ
first, and upon moving and upon being moved, clear, and that the
κίνησις οὐκ ἐν τῷ εἴδει ἀλλ' ἐν τῷ κινουμένῳ καὶ κινητῷ κατ'
motion not in the form but in the moving and movable according to
ἐνέργειαν. ἡ μὲν οὖν κατὰ συμβεβηκὸς μεταβολὴ
activity. the on one hand therefore according to contingent change
ἀφείσθω· ἐν ἅπασί τε γάρ ἐστι καὶ αἰεὶ καὶ πάντων· ἡ
let it be disregarded; in all and for it is and always and of all; the
δὲ μὴ κατὰ συμβεβηκὸς οὐκ ἐν ἅπασιν, ἀλλ' ἐν τοῖς ἐναντίοις
but not according to contingent not in all, but in the opposites
καὶ τοῖς μεταξὺ καὶ ἐν ἀντιφάσει· τούτου δὲ πίστις
and the in between and in contradiction; of this on the other hand trust
ἐκ τῆς ἐπαγωγῆς. ἐκ δὲ τοῦ μεταξὺ μετα βάλλει·
from the induction. from on the other hand the in between changes it throws;
χρῆται γὰρ αὐτῷ ὡς ἐναντίῳ ὄντι πρὸς ἑκάτερον· ἔστι γάρ πως
it uses for it as opposite being towards each it is for somehow

τὸ μεταξὺ τὰ ἄκρα. διὸ καὶ τοῦτο πρὸς ἐκεῖνα κἀκεῖνα πρὸς
the between the extremes. therefore and this towards those and those towards
τοῦτο λέγεταί πως ἐναντία, οἷον ἡ μέση ὀξεῖα πρὸς τὴν ὑπάτην
this is called somehow opposite, such as the middle sharp towards the highest
καὶ βαρεῖα πρὸς τὴν νητήν, καὶ τὸ φαιὸν λευκὸν πρὸς τὸ μέλαν
and heavy towards the lowest, and the gray white towards the black
καὶ μέλαν πρὸς τὸ λευκόν.
and black towards the white.

ἐπεὶ δὲ [225a] πᾶσα μεταβολή ἐστιν ἔκ τινος εἴς τι (δηλοῖ
since but [225a] every change is from someone into something (indicates
δὲ καὶ τοὔνομα· μετ' ἄλλο γάρ τι καὶ τὸ μὲν πρότερον
but and the name· after another for something and the on one hand earlier
δηλοῖ, τὸ δ' ὕστερον), μεταβάλλοι ἂν τὸ μεταβάλλον
indicates, the on the other hand later), might change would the changing
τετραχῶς· ἢ γὰρ ἐξ ὑποκειμένου εἰς ὑποκείμενον, ἢ ἐξ ὑποκειμένου
in four ways· or for from subject into subject, or from subject
εἰς μὴ ὑποκείμενον, ἢ οὐκ ἐξ ὑποκειμένου εἰς ὑποκείμενον, ἢ οὐκ
into not subject, or not from subject into subject, or not
ἐξ ὑποκειμένου εἰς μὴ ὑποκείμενον· λέγω δὲ ὑποκείμενον τὸ
from subject into not subject· I say but subject the
καταφάσει δηλούμενον. ὕστε ἀνάγκη ἐκ τῶν εἰρημένων τρεῖς
namely being indicated. later necessity from the previously stated three
εἶναι μεταβολάς, τήν τε ἐξ ὑποκειμένου εἰς ὑποκείμενον, καὶ τὴν
to be changes, the and from subject into subject, and the
ἐξ ὑποκειμένου εἰς μὴ ὑποκείμενον, καὶ τὴν ἐκ μὴ ὑποκειμένου
from subject into not subject, and the from not subject

εἰς ὑποκείμενον. ἡ γὰρ οὐκ ἐξ ὑποκειμένου εἰς μὴ ὑποκείμενον
into subject. the for not out of subject into not subject
οὐκ ἔστιν μεταβολὴ διὰ τὸ μὴ εἶναι κατ' ἀντίθεσιν· οὔτε
not is change through the not to be according to opposition; neither
γὰρ ἐναντία οὔτε ἀντίφασίς ἐστιν.
for contrary nor contradiction is.

ἡ μὲν οὖν οὐκ ἐξ ὑποκειμένου εἰς ὑποκείμενον μεταβολὴ
the indeed therefore not out of the substrate into substratum change
κατ' ἀντίφασιν γένεσίς ἐστιν, ἡ μὲν ἁπλῶς ἁπλῆ, ἡ δὲ τὶς
according to contradiction generation is, the indeed simply simple, the but some
τινός (οἷον ἡ μὲν ἐκ μὴ λευκοῦ εἰς λευκὸν γένεσις τούτου, ἡ
certain (such as the indeed from not white into whiteness generation of this, the
δ' ἐκ τοῦ μὴ ὄντος ἁπλῶς εἰς οὐσίαν γένεσις ἁπλῶς, καθ'
but from the not being simply into essence generation simply, according to
ἣν ἁπλῶς γίγνεσθαι καὶ οὐ τὶ γίγνεσθαι λέγομεν)· ἡ δ'
which simply to become and not anything to become we say). the but
ἐξ ὑποκειμένου εἰς οὐχ ὑποκεί μενον φθορά, ἁπλῶς μὲν ἡ
out of the substrate into not substratum substratum decay, simply indeed the
ἐκ τῆς οὐσίας εἰς τὸ μὴ εἶναι, τὶς δὲ ἡ εἰς τὴν ἀντικειμένην
from the essence into the not being, some but the into the opposing
ἀπόφασιν, καθάπερ ἐλέχθη καὶ ἐπὶ τῆς γενέσεως.
assertion, as it was said and regarding of the generation.

εἰ δὴ τὸ μὴ ὂν λέγεται πλεοναχῶς, καὶ μήτε τὸ κατὰ
if indeed the not being is said redundantly, and neither the according to
σύνθεσιν ἢ διαίρεσιν ἐνδέχεται κινεῖσθαι μήτε τὸ κατὰ δύναμιν,
synthesis or division may be moved nor the according to power,

τὸ τῷ ἁπλῶς κατ' ἐνέργειαν ὄντι ἀντικείμενον (τὸ μὲν γὰρ
the to the simply by actualization being object (the on one hand for
μὴ λευκὸν ἢ μὴ ἀγαθὸν ὅμως ἐνδέχεται κινεῖσθαι κατὰ
not white or not good nevertheless may be moved according to
συμβεβηκός, εἴη γὰρ <ἂν> ἄνθρωπος τὸ μὴ λευκόν· τὸ δ'
contingent, let it be for <if> a person the not white; the but
ἁπλῶς μὴ τόδε οὐδαμῶς), ἀδύνατον [γὰρ] τὸ μὴ ὂν κινεῖσθαι
simply not this in no way), impossible [for] the not being to be moved
(εἰ δὲ τοῦτο, καὶ τὴν γένεσιν κίνησιν εἶναι· γίγνεται γὰρ τὸ
(if but this, and the generation movement to be; it happens for the
μὴ ὄν· εἰ γὰρ καὶ ὅτι μάλιστα κατὰ συμβεβηκὸς γίγνεται,
not being; if for and that especially according to having happened becomes,
ἀλλ' ὅμως ἀληθὲς εἰπεῖν ὅτι ὑπάρχει τὸ μὴ ὂν κατὰ τοῦ
but nevertheless true to say that exists the not being according to the
γιγνομένου ἁπλῶς)-ὁμοίως δὲ καὶ τὸ ἠρεμεῖν.
happening simply)-likewise and also the tranquility.

ταῦτά τε δὴ συμβαίνει δυσχερῆ [τῷ κινεῖσθαι τὸ μὴ ὄν] καὶ
these and indeed happens difficulties to the being moved the not being] and
εἰ πᾶν τὸ κινούμενον ἐν τόπῳ, τὸ δὲ μὴ ὂν οὐκ ἔστιν ἐν
if everything the moving in place, the but not being is not is in
τόπῳ· εἴη γὰρ ἄν που.
place. let it be for perhaps somewhere.

οὐδὲ δὴ ἡ φθορὰ κίνησις· ἐναντίον μὲν γὰρ κινήσει ἢ
nor indeed the decay motion; contrary on one hand for motion or
κίνησις ἢ ἠρεμία, ἡ δὲ φθορὰ γενέσει ἐναντίον. ἐπεὶ δὲ πᾶσα
movement or rest, the but decay by generation contrary. since and every

κίνησις μεταβολή τις, μεταβολαὶ δὲ τρεῖς αἱ εἰρημέναι, τούτων
motion change some, changes and three the mentioned, of these
δὲ αἱ κατὰ γένεσιν καὶ φθορὰν [225b] οὐ κινήσεις, αὗται δ'
and the according to generation and decay [225b] not changes, these but
εἰσὶν αἱ κατ' ἀντίφασιν, ἀνάγκη τὴν ἐξ ὑποκειμένου εἰς
are the according to contradiction, necessity the from substrate into
ὑποκείμενον μεταβολὴν κίνησιν εἶναι μόνην. τὰ δ' ὑποκείμενα ἢ
subject change motion to be alone. the and substrates or
ἐναντία ἢ μεταξύ (καὶ γὰρ ἡ στέρησις κείσθω ἐναντίον), καὶ
contrary or between (and for the deprivation let it be contrary), and
δηλοῦται καταφάσει, τὸ γυμνὸν καὶ νωδὸν καὶ μέλαν. εἰ οὖν
is indicated by assertion, the naked and dull and dark. if therefore
αἱ κατηγορίαι διήρηνται οὐσίᾳ καὶ ποιότητι καὶ τῷ ποὺ [καὶ
the categories are divided by substance and quality and to the where [and
τῷ ποτὲ] καὶ τῷ πρός τι καὶ τῷ ποσῷ καὶ τῷ ποιεῖν
to the when] and to the towards something and to the quantity and to the doing
ἢ πάσχειν, ἀνάγκη τρεῖς εἶναι κινήσεις, τήν τε τοῦ ποιοῦ καὶ τὴν
or suffering, necessity three to be motions, the and of the doer and the
τοῦ ποσοῦ καὶ τὴν κατὰ τόπον.
of the quantity and the according to place.

Κεφάλαιο 2

Chapter 2

Κατ' οὐσίαν δ' οὐκ ἔστιν κίνησις διὰ τὸ μηδὲν εἶναι οὐσίᾳ
According to essence and not is motion through the nothing to be essence
τῶν ὄντων ἐναντίον. οὐδὲ δὴ τοῦ πρός τι· ἐνδέχεται
of the beings contrary. nor indeed of the towards something; it is possible

γὰρ θατέρου μεταβάλλοντος <ἀληθεύεσθαι καὶ μὴ> ἀληθεύεσθαι

for the other changing to be true and not> to be true

θάτερον μηδὲν μεταβάλλον, ὥστε κατὰ συμβεβηκὸς ἡ

another nothing changing, so that according to what has happened the

κίνησις αὐτῶν. οὐδὲ δὴ ποιοῦντος καὶ πάσχοντος, ἢ κινουμένου

motion of them. nor indeed of making and of suffering, or of moving

καὶ κινοῦντος, ὅτι οὐκ ἔστι κινήσεως κίνησις οὐδὲ γενέσεως

and of moving, because not is of motion motion nor of generation

γένεσις, οὐδ' ὅλως μεταβολῆς μεταβολή.

generation, nor altogether of change change.

πρῶτον μὲν γὰρ διχῶς ἐνδέχεται κινήσεως εἶναι κίνησιν, ἢ ὡς

First indeed for in two ways it is possible of movement to be a movement, or as

ὑποκειμένου (οἷον ἄνθρωπος κινεῖται ὅτι ἐκ λευκοῦ εἰς μέλαν

a subject (such as a human moves that from white to black

μεταβάλλει· ἆρά γε οὕτω καὶ ἡ κίνησις ἢ θερμαίνεται ἢ

it changes. Is it then really thus and the movement or is heated or

ψύχεται ἢ τόπον ἀλλάττει ἢ αὐξάνεται ἢ φθίνει; τοῦτο δὲ

is cooled or changes place changes or increases or decreases? This but

ἀδύνατον· οὐ γὰρ τῶν ὑποκειμένων τι ἡ μεταβολή), ἢ τῷ

is impossible. Not for of the subjects anything the change), or to the

ἕτερόν τι ὑποκείμενον ἐκ μεταβολῆς μεταβάλλειν εἰς ἕτερον

other something subject from change to change into another

εἶδος [οἷον ἄνθρωπος ἐκ νόσου εἰς ὑγίειαν]. ἀλλ' οὐδὲ τοῦτο

form [such as a human from disease to health]. But neither this

δυνατὸν πλὴν κατὰ συμβεβηκός· αὐτὴ γὰρ ἡ κίνησις

is possible except according to contingent circumstances. Itself for the movement

ἐξ ἄλλου εἴδους εἰς ἄλλο ἐστὶ μεταβολή <οἷον ἀνθρώπου ἐκ
from another form to another is change such as of a human out of
νόσου εἰς ὑγίειαν> · καὶ ἡ γένεσις δὲ καὶ ἡ φθορὰ ὡσαύτως,
disease into health. And the generation but and the decay likewise,
πλὴν αἱ μὲν εἰς ἀντικείμενα ὡδί, ἡ δὲ ὡδί, ἡ κίνησις.
except the on one hand into opposites thus, the but thus, the motion.
ἅμα οὖν μεταβάλλει ἐξ ὑγιείας εἰς νόσον καὶ ἐξ αὐτῆς
Together therefore changes from health into disease and from it
ταύτης τῆς μεταβολῆς εἰς ἄλλην. δῆλον δὴ ὅτι ὅταν νοσήσῃ,
this the change into another. clear indeed that when he/she falls ill,
μεταβεβληκὸς ἔσται εἰς ὁποιανοῦν (ἐνδέχεται γὰρ ἠρεμεῖν),
having changed, he/she will be into any (may indeed be at rest),
καὶ ἔτι εἰς μὴ τὴν τυχοῦσαν αἰεί, κἀκείνη ἔκ τινος εἴς τι
and yet into not the happening always, and that out of something into anything
ἕτερον ἔσται, ὥστε καὶ ἡ ἀντι κειμένη ἔσται ὑγίανσις · ἀλλὰ
different will be, so that also the opposite will be will be health. But
τῷ συμβεβηκέναι, οἷον ἐξ ἀναμνήσεως εἰς λήθην μεταβάλλει,
by the happening, such as from memory into forgetfulness changes,
ὅτι ᾧ ὑπάρχει, ἐκεῖνο μεταβάλλει ὁτὲ μὲν εἰς
because to whom it exists, that changes sometimes on the one hand into
ἐπιστήμην ὁτὲ δ' εἰς ἄγνοιαν.
knowledge sometimes but into ignorance.

ἔτι εἰς ἄπειρον βαδιεῖται, εἰ ἔσται μεταβολῆς μεταβολὴ καὶ
still into the infinite it is going, if it will be of change change and
γενέσεως γένεσις. ἀνάγκη δὴ καὶ τὴν προτέραν, εἰ ἡ ὑστέρα
of becoming becoming. necessity indeed and the previous, if the latter

[226a] ἔσται, οἷον εἰ ἡ ἁπλῆ γένεσις ἐγίγνετό ποτε, καὶ

[226a] it will be, such as if the simple becoming happened at some time, and

τὸ γιγνόμενον ἐγίγνετο, ὥστε οὔπω ἦν τὸ γιγνόμενον ἁπλῶς,

the becoming happened, so that not yet it was the becoming simply,

ἀλλά τι γιγνόμενον γιγνόμενον ἤδη, καὶ πάλιν τοῦτ' ἐγίγνετό

but something becoming becoming, already, and again this happened

ποτε, ὥστ' οὐκ ἦν πω τότε γιγνόμενον γιγνόμενον. ἐπεὶ

at some time, so that not it was yet then becoming becoming. since

δὲ τῶν ἀπείρων οὐκ ἔστιν τι πρῶτον, οὐκ ἔσται τὸ

but of the infinite there is not there is anything first, not it will be the

πρῶτον, ὥστ' οὐδὲ τὸ ἐχόμενον· οὔτε γί γνεσθαι οὖν

first, so that not even the subsequent· neither is to be generated therefore

οὔτε κινεῖσθαι οἷόν τε οὔτε μεταβάλλειν οὐδέν.

nor to be moved such and nor to change nothing.

ἔτι τοῦ αὐτοῦ κίνησις ἡ ἐναντία (καὶ ἔτι ἠρέμησις), καὶ γένεσις

yet of the same motion the opposite (and yet rest), and generation

καὶ φθορά, ὥστε τὸ γιγνόμενον γιγνόμενον ὅταν γένηται

and decay, so that the becoming becoming whenever it comes to be

γιγνόμενον, τότε φθείρεται· οὔτε γὰρ εὐθὺς γιγνόμενον οὔθ'

becoming, then it is destroyed. neither for immediately becoming nor

ὕστερον· εἶναι γὰρ δεῖ τὸ φθειρόμενον.

later. to be for it is necessary the destroyed.

ἔτι ὕλην δεῖ ὑπεῖναι καὶ τῷ γιγνομένῳ καὶ τῷ μεταβάλλοντι. τίς οὖν

still matter must exist and the becoming and the changing. who then

ἔσται—ὥσπερ τὸ ἀλλοιωτὸν σῶμα ἢ ψυχή, οὕτω τί τὸ γιγνόμενον

will be-as the alterable body or soul, so what the becoming

κίνησις ἢ γένεσις; καὶ πάλιν τί εἰς ὃ κινοῦνται; δεῖ γὰρ εἶναι

motion or generation? and again what into what do they move? must for be

[τι] τὴν τοῦδε ἐκ τοῦδε εἰς τόδε κίνησιν [καὶ μὴ κίνησιν] ἢ

[something] the of this from this into this motion [and not motion] or

γένεσιν. ἅμα δὲ πῶς καὶ ἔσται; οὐ γὰρ ἔσται μάθησις ἡ

generation. at once but how and will it be? not for it will be learning the

τῆς μαθήσεως γένεσις, ὥστ' οὐδὲ γενέσεως γένεσις, οὐδέ τις

of the learning generation, so that neither of generation generation, nor anyone

τινός.

of someone.

ἔτι εἰ τρία εἴδη κινήσεώς ἐστιν, τούτων τινὰ ἀνάγκη εἶναι καὶ

still if three kinds of motion there are, of these some necessity to be and

τὴν ὑποκειμένην φύσιν καὶ εἰς ἃ κινοῦνται, οἷον τὴν φορὰν

the underlying nature and into that they are moved, such as the motion

ἀλλοιοῦσθαι ἢ φέρεσθαι.

to be altered or to be carried.

ὅλως δὲ ἐπεὶ κινεῖται [τὸ κινούμενον] πᾶν τριχῶς, ἢ κατὰ

altogether but since it is moved the moving all in three ways, or according to

συμβεβηκὸς ἢ τῷ μέρος τι ἢ [τῷ] καθ' αὑτό, κατὰ

accident or by the part something or by the in respect to itself, according to

συμβεβηκὸς μόνον ἂν ἐνδέχοιτο μεταβάλλειν τὴν μεταβολήν,

accident only if it were to happen to change the change,

οἷον εἰ ὁ ὑγιαζόμενος τρέχοι ἢ μανθάνοι· τὴν δὲ κατὰ

such as if the healthy were to run or learn; the but according to

συμβεβηκὸς ἀφεῖμεν πάλαι.

accident we leave off long ago.

ἐπεὶ δὲ οὔτε οὐσίας οὔτε τοῦ πρός τι οὔτε τοῦ ποιεῖν καὶ
since but neither substance nor of the towards something nor of the causing and
πάσχειν, λείπεται κατὰ τὸ ποιὸν καὶ τὸ ποσὸν καὶ τὸ ποὺ
suffering, it is left according to the quality and the quantity and the where
κίνησιν εἶναι μόνον· ἐν ἑκάστῳ γὰρ ἔστι τούτων ἐναντίωσις. ἡ
movement to be only; in each for there is of these opposition. the
μὲν οὖν κατὰ τὸ ποιὸν κίνησις ἀλλοίωσις ἔστω·
on the one hand therefore according to the quality movement alteration let it be;
τοῦτο γὰρ ἐπέζευκται κοινὸν ὄνομα. λέγω δὲ τὸ ποιὸν οὐ τὸ ἐν τῇ
this for is associated common name. I say but the quality not the in the
οὐσίᾳ (καὶ γὰρ ἡ διαφορὰ ποιότης) ἀλλὰ τὸ παθητικόν, καθ'
substance (and for the difference quality) but the passive, according to
ὃ λέγεται πάσχειν ἢ ἀπαθὲς εἶναι. ἡ δὲ κατὰ τὸ ποσὸν
which is called suffering or impassive to be. the but according to the quantity
τὸ μὲν κοινὸν ἀνώνυμος, καθ' ἑκάτερον δ' αὔξησις
the on the one hand common anonymous, according to each of increase
καὶ φθίσις, ἡ μὲν εἰς τὸ τέλειον μέγεθος αὔξησις, ἡ δ' ἐκ
and decrease, the one into the perfect magnitude, increase, the other from
τούτου φθίσις. ἡ δὲ κατὰ τόπον καὶ τὸ κοινὸν καὶ τὸ ἴδιον
this decrease. the but according to place and the common and the private
ἀνώνυμος, ἔστω δὲ φορὰ καλουμένη τὸ κοινόν· καίτοι λέγεταί
anonymous, let it be but a movement called the common. And yet it is said
γε ταῦτα φέρεσθαι μόνα κυρίως, ὅταν μὴ ἐφ' αὑτοῖς ᾖ τὸ
indeed these to be carried alone chiefly, when not upon themselves it is the
στῆναι τοῖς μεταβάλλουσι τὸν τόπον, καὶ ὅσα [226b] μὴ αὐτὰ
standing to those changing the place, and as many as [226b] not they

ἑαυτὰ κινεῖ κατὰ τόπον.

themselves move according to place.

ἡ δ' ἐν τῷ αὐτῷ εἴδει μεταβολὴ ἐπὶ τὸ μᾶλλον καὶ ἧττον ἀλλοίωσίς

the and in the same form change to the more and less alteration

ἐστιν· ἢ γὰρ ἐξ ἐναντίου ἢ εἰς ἐναντίον κίνησίς ἐστιν, ἢ ἁπλῶς

is. or for from opposite or to opposite movement is, or simply

ἢ πή· ἐπὶ μὲν γὰρ τὸ ἧττον ἰοῦσα εἰς τοὐναντίον λεχθήσεται

or else. to on the one hand for the less going to the opposite will be said

μεταβάλλειν, ἐπὶ δὲ τὸ μᾶλλον ὡς ἐκ τοὐναντίου εἰς αὐτό. διαφέρει

to change, to but the more as from the opposite to itself. differs

γὰρ οὐδὲν πῇ μεταβάλλειν ἢ ἁπλῶς, πλὴν πῇ δεήσει τἀναντία

for nothing at all to change or simply, except that it would need the opposites

ὑπάρχειν· τὸ δὲ μᾶλλον καὶ ἧττόν ἐστι τὸ πλέον ἢ ἔλαττον

to exist. the but more and less is the greater than lesser

ἐνυπάρχειν τοῦ ἐναντίου καὶ μή. ὅτι μὲν οὖν αὗται τρεῖς

to exist in the opposite and not. that on one hand therefore these three

μόναι κινήσεις εἰσίν, ἐκ τούτων δῆλον·

only movements are, from these clear:

ἀκίνητον δ' ἐστὶ τό τε ὅλως ἀδύνατον κινηθῆναι, ὥσπερ ὁ

immovable but is the and altogether impossible to be moved, just as the

ψόφος ἀόρατος, καὶ τὸ ἐν πολλῷ χρόνῳ μόλις κινούμενον ἢ

sound is invisible, and the in much time hardly being moved or

τὸ βραδέως ἀρχόμενον, ὃ λέγεται δυσκίνητον, καὶ τὸ

the slowly starting, which is said to be difficult to move, and the

πεφυκὸς μὲν κινεῖσθαι καὶ δυνάμενον, μὴ κινούμενον δὲ τότε

natural indeed to be moved and capable, not being moved but then

ὅτε πέφυκε καὶ οὗ καὶ ὡς, ὅπερ ἠρεμεῖν καλῶ τῶν

when it has come to be and of which also as, which rests I call of the

ἀκινήτων μόνον· ἐναντίον γὰρ ἠρεμία κινήσει, ὥστε στέρησις

immovable things only: in contrast for rest sets in motion, so that deprivation

ἂν εἴη τοῦ δεκτικοῦ.

would be of the recipient.

τί μὲν οὖν ἐστι κίνησις καὶ τί ἠρεμία, καὶ πόσαι μεταβολαὶ καὶ

What then therefore is motion and what rest, and how many changes and

ποῖαι κινήσεις, φανερὸν ἐκ τῶν εἰρημένων.

what kinds of motions, it is evident from the aforementioned.

Κεφάλαιο 3

Chapter 3

Μετὰ δὲ ταῦτα λέγωμεν τί ἐστιν τὸ ἅμα καὶ χωρίς, καὶ τί

After but these let us say what is the at once and separately, and what

τὸ ἅπτεσθαι, καὶ τί τὸ μεταξὺ καὶ τί τὸ ἐφεξῆς καὶ τί τὸ

the to touch, and what the in between and what the following and what the

ἐχόμενον καὶ συνεχές, καὶ τοῖς ποίοις ἕκαστον τούτων ὑπάρχειν

holding and continuous, and to the kinds each of these to exist

πέφυκεν.

is destined.

ἅμα μὲν οὖν λέγω ταῦτ' εἶναι κατὰ τόπον, ὅσα

At once on the one hand therefore I say these to be according to place, as many as

ἐν ἑνὶ τόπῳ ἐστὶ πρώτῳ, χωρὶς δὲ ὅσα ἐν ἑτέρῳ,

in one place is first, separately but as many as in another,

ἅπτεσθαι δὲ ὧν τὰ ἄκρα ἅμα.

to touch and of which the extremes at once.

μεταξὺ δὲ εἰς ὃ πέφυκε πρότερον ἀφικνεῖσθαι τὸ μεταβάλλον ἢ

between and into that has come to be previously to arrive the changing or

εἰς ὃ ἔσχατον μεταβάλλει κατὰ φύσιν συνεχῶς μεταβάλλον.

into that last changes according to nature continuously changing.

[ἐν ἐλαχίστοις δ' ἐστι τὸ μεταξὺ τρισίν· ἔσχατον μὲν γάρ ἐστι

in the slightest but is the between three: last indeed for is

τῆς μεταβολῆς τὸ εναντιον.] συνεχῶς δὲ κινεῖται τὸ μηθὲν ἢ ὅτι

of the change the opposite.] continuously and moves the nothing or that

ὀλίγιστον διαλεῖπον τοῦ πράγματος—μὴ τοῦ χρόνου (οὐδὲν γὰρ

slight leaving out of the thing-not of time nothing for

κωλύει διαλείποντα, καὶ εὐθὺς δὲ μετὰ τὴν ὑπάτην φθέγξασθαι

hinders to leave out, and immediately and after the supreme to sound

τὴν νεάτην) ἀλλὰ τοῦ πράγματος ἐν ᾧ κινεῖται. [<ἐπεὶ δὲ πᾶσα

the new) but of the thing in which it moves. [since and every

μεταβολὴ ἐν τοῖς ἀντικειμένοις, τὰ δ' ἀντικείμενα τά τε ἐναντία καὶ

change in the opposites, the but opposites the and contrary and

τὰ κατὰ ἀντίφασιν, ἀντιφάσεως δ' οὐδὲν ἀνὰ μέσον, φανερὸν

the according to contradiction, of contradiction but nothing in the middle, it is clear

ὅτι ἐν τοῖς ἐναντίοις ἔσται τὸ μεταξύ. [226c] ἐν ἐλαχίστοις δ'

that in the opposites it will be the between. [226c] in the least but

ἐστὶ τὸ μεταξὺ τρισίν· ἔσχατον μὲν γάρ ἐστι τῆς μεταβολῆς

it is the between three: the last on one hand for it is of the change

τὸ ἐναντίον,>] τοῦτο δὲ ἔν τε ταῖς κατὰ τόπον καὶ ἐν ταῖς

the opposite,> this but in and the according to place and in the

ἄλλαις μεταβολαῖς φανερόν. ἐναντίον δὲ κατὰ τόπον τὸ

other changes it is clear. opposite but according to place the

κατ' εὐθεῖαν ἀπέχον πλεῖστον· ἡ γὰρ ἐλαχίστη πεπέρανται,

according to straight farthest most; the for least are separated,

μέτρον δὲ τὸ πεπερασμένον.

the measure but the finite.

ἐφεξῆς δὲ οὗ μετὰ τὴν ἀρχὴν ὄντος ἢ θέσει ἢ εἴδει ἢ ἄλλῳ

subsequently and of which after the beginning being or position or type or another

τινὶ οὕτως ἀφορισθέντος [227a] μηδὲν μεταξύ ἐστι τῶν ἐν ταὐτῷ

some thus having been defined [227a] nothing between is of the in the same

γένει καὶ οὗ ἐφεξῆς ἐστιν (λέγω δ' οἷον γραμμὴ γραμμῆς

genus and of which subsequently it is (I say and such as a line of a line

ἢ γραμμαί, ἢ μονάδος μονὰς ἢ μονάδες, ἢ οἰκίας οἰκία· ἄλλο

or letters, or of a unit units or units, or of a house house. another

δ' οὐδὲν κωλύει μεταξὺ εἶναι). τὸ γὰρ ἐφεξῆς τινὶ ἐφεξῆς

and nothing hinders between to be). the for subsequently to some subsequently

καὶ ὕστερόν τι· οὐ γὰρ τὸ ἓν ἐφεξῆς τοῖν δυοῖν, οὐδ' ἡ

and later something; not for the one subsequently of the two two, nor the

νουμηνία τῇ δευτέρᾳ ἐφεξῆς, ἀλλὰ ταῦτ' ἐκείνοις.

new moon in the second subsequently, but these to those.

ἐχόμενον δὲ ὃ ἂν ἐφεξῆς ὂν ἅπτηται. τὸ δὲ συνεχὲς

that is held and whatever that may subsequently being is touched. the but continuous

ἔστι μὲν ὅπερ ἐχόμενόν τι, λέγω δ' εἶναι συνεχὲς ὅταν

it is indeed which holds something, I say but to be continuous when

ταὐτὸ γένηται καὶ ἓν τὸ ἑκατέρου πέρας οἷς ἅπτονται, καὶ

the same happens and one the of each end to which they attach, and

ὥσπερ σημαίνει τοὔνομα, συνέχηται. τοῦτο δ' οὐχ οἷόν τε
just as it signifies the name, it is held together. this but not possible or
δυοῖν ὄντοιν εἶναι τοῖν ἐσχάτοιν. τούτου δὲ διωρισμένου φανερὸν
of two beings to be of the last. of this but having been defined evident
ὅτι ἐν τούτοις ἐστὶ τὸ συνεχές, ἐξ ὧν ἕν τι πέφυκε
that in these there is the continuous, from which one something has come to be
γίγνεσθαι κατὰ τὴν σύναψιν. καὶ ὥς ποτε γίγνεται τὸ συνέχον
to become according to the connection. and as ever becomes the holding
ἕν, οὕτω καὶ τὸ ὅλον ἔσται ἕν, οἷον ἢ γόμφῳ ἢ κόλλῃ ἢ
one, so and the whole will be one, such as either by a joint or by a glue or
ἁφῇ ἢ προσφύσει.
by a touch or by a fitting.

φανερὸν δὲ καὶ ὅτι πρῶτον τὸ ἐφεξῆς ἐστι· τὸ μὲν γὰρ ἁπτόμενον
evident but and that first the subsequently is. the indeed for touching
ἐφεξῆς ἀνάγκη εἶναι, τὸ δ' ἐφεξῆς οὐ πᾶν ἅπτεσθαι (διὸ
subsequently necessity to be, the but subsequently not all to touch (therefore
καὶ ἐν προτέροις τῷ λόγῳ τὸ ἐφεξῆς ἔστιν, οἷον ἐν ἀριθμοῖς,
and in previous the discourse the subsequently is, such as in numbers,
ἁφὴ δ' οὐκ ἔστιν), καὶ εἰ μὲν συνεχές, ἀνάγκη ἅπτεσθαι, εἰ δ'
contact but not is), and if indeed continuous, necessity to touch, if but
ἅπτεται, οὔπω συνεχές· οὐ γὰρ ἀνάγκη ἓν εἶναι αὐτῶν τὰ ἄκρα,
it touches, not yet continuous. not for necessity one to be of them the extremes,
εἰ ἅμα εἶεν· ἀλλ' εἰ ἕν, ἀνάγκη καὶ ἅμα. ὥστε ἡ σύμφυσις
if together they are; but if one, necessity and together. thus the nature
ὑστάτη κατὰ τὴν γένεσιν· ἀνάγκη γὰρ ἅψασθαι εἰ συμφύσεται
last according to the genesis. necessity for to touch if it joins

τὰ ἄκρα, τὰ δὲ ἁπτόμενα οὐ πάντα συμπέφυκεν· ἐν οἷς δὲ μὴ
the extremes, the and touching not all has been mixed in which but not
ἔστιν ἁφή, δῆλον ὅτι οὐκ ἔστιν οὐδὲ σύμφυσις ἐν τούτοις.
is contact, it is clear that not is nor unity in these.
ὥστ' εἰ ἔστι στιγμὴ καὶ μονὰς οἵας λέγουσι κεχωρισμένας, οὐχ
Therefore if it is a point and a unit what they say separated, not
οἷόν τε εἶναι μονάδα καὶ στιγμὴν τὸ αὐτό· ταῖς μὲν γὰρ
possible and to be a unit and a point the same; for the indeed for
ὑπάρχει τὸ ἅπτεσθαι, ταῖς δὲ μονάσιν τὸ ἐφεξῆς, καὶ τῶν μὲν
exists the touching, for the but units the following, and of the indeed
ἐνδέχεται εἶναί τι μεταξύ (πᾶσα γὰρ γραμμὴ μεταξὺ στιγμῶν),
it is possible to be something between (every for line between points),
τῶν δ' οὐκ ἀνάγκη· οὐδὲ γὰρ μεταξὺ δυάδος καὶ μονάδος.
of the and not necessary; nor for between a duality and a unity.

τί μὲν οὖν ἐστι τὸ ἅμα καὶ χωρίς, καὶ τί τὸ ἅπτεσθαι,
what on the one hand therefore is the at once and apart, and what the to be touched,
καὶ τί τὸ μεταξὺ [227b] καὶ τὸ ἐφεξῆς, καὶ τί τὸ ἐχόμενον
and what the in between [227b] and the following, and what the holding
καὶ τὸ συνεχές, καὶ τοῖς ποίοις ἕκαστον τούτων ὑπάρχει,
and the continuous, and to those of what sort each of these exists,
εἴρηται. Μία δὲ κίνησις λέγεται πολλαχῶς· τὸ γὰρ ἓν
it has been said. One but movement is called in many ways; the for one
πολλαχῶς λέγομεν.
in many ways we say.

Κεφάλαιο 4

Chapter 4

γένει μὲν οὖν μία κατὰ τὰ σχήματα τῆς κατηγορίας

of kinds on the one hand therefore one according to the forms of the category

ἐστί

is

(φορὰ μὲν γὰρ πάσῃ φορᾷ τῷ γένει μία, ἀλλοίωσις

(of occurrence on the one hand for in every occurrence to the kind one, alteration

δὲ φορᾶς ἑτέρα τῷ γένει),

but of occurrence another to the kind),

εἴδει δὲ μία, ὅταν τῷ γένει μία οὖσα καὶ ἐν ἀτόμῳ εἴδει ᾖ. οἷον

form but one, when to the nature one being and in atom form it is. such as

χρώματος μὲν εἰσὶ διαφοραί—τοιγαροῦν ἄλλη τῷ εἴδει

color on the one hand there are differences-therefore another to the form

μέλανσις καὶ λεύκανσις [πᾶσα οὖν λεύκανσις πάσῃ λευκάνσει ἡ

blackness and whiteness [all thus whiteness in all whiteness the

αὐτὴ κατ' εἶδος ἔσται καὶ πᾶσα μέλανσις μελάνσει]—λευκότητος

same according to form will be and all blackness blackness]-of whiteness

δ' οὐκέτι· διὸ τῷ εἴδει μία λεύκανσις λευκάνσει πάσῃ. εἰ δ'

but no longer; therefore to the form one whiteness whiteness in all. if but

ἔστιν ἄτθ' ἃ καὶ γένη ἅμα καὶ εἴδη ἐστίν, δῆλον ὡς ἔστιν

there is among those and kinds together and forms there are, clear that there is

ὡς εἴδει μία ἔσται, ἁπλῶς δὲ μία εἴδει οὔ, οἷον ἡ μάθησις, εἰ ἡ

as form one will be, simply but one form no, such as the learning, if the

ἐπιστήμη εἶδος μὲν ὑπολήψεως, γένος δὲ τῶν ἐπιστημῶν.

science form on the one hand of hypothesis, the kind but of the sciences.

ἀπορήσειε δ' ἄν τις εἰ εἴδει μία <ἡ> κίνησις, ὅταν ἐκ

he would wonder but if someone whether he knows one the motion, when from

τοῦ αὐτοῦ τὸ αὐτὸ εἰς τὸ αὐτὸ μεταβάλλῃ, οἷον ἡ μία στιγμὴ ἐκ
the same the same to the same changes, such as the one moment from
τοῦδε τοῦ τόπου εἰς τόνδε τὸν τόπον πάλιν καὶ πάλιν. εἰ δὲ τοῦτ',
this the place to that the place again and again. if but this
ἔσται ἡ κυκλοφορία τῇ εὐθυφορίᾳ ἡ αὐτὴ καὶ ἡ κύλισις τῇ
will be the circulation in the straight course the same and the rolling in the
βαδίσει. ἢ διώρισται, τὸ ἐν ᾧ ἂν ἕτερον ᾖ τῷ
walking. Or it has been defined, the in which one whichever other it may be in the
εἴδει, ὅτι ἑτέρα ἡ κίνησις, τὸ δὲ περιφερὲς τοῦ εὐθέος ἕτερον
form, that another the motion, the and the circular of the straight another
τῷ εἴδει; γένει μὲν οὖν καὶ εἴδει κίνησις μία οὕτως,
in the form? In terms on one hand therefore and forms motion one thus,

ἁπλῶς δὲ μία κίνησις ἡ τῇ οὐσίᾳ μία καὶ τῷ ἀριθμῷ· τίς δ'
simply but one motion the in essence one and with number. who but
ἡ τοιαύτη, δῆλον διελομένοις. τρία γάρ ἐστι τὸν ἀριθμὸν
the such, clear to those distinguishing. three for it is the number
περὶ ἃ λέγομεν τὴν κίνησιν, ὃ καὶ ἐν ᾧ καὶ ὅτε.
concerning which we are talking about the motion, which and in which and when.
λέγω δ' ὅτι ἀνάγκη εἶναί τι τὸ κινούμενον, οἷον ἄνθρωπον
I say but that necessity to be something the moving, such as a person
ἢ χρυσόν, καὶ ἔν τινι τοῦτο κινεῖσθαι, οἷον ἐν τόπῳ ἢ ἐν
or gold, and in something this to be moved, such as in a place or in
πάθει, καὶ ποτέ· ἐν χρόνῳ γὰρ πᾶν κινεῖται. τούτων δὲ
an affection, and at some time; in time for everything is moved. of these but
τὸ μὲν εἶναι τῷ γένει ἢ τῷ εἴδει μίαν ἐστὶν ἐν τῷ πράγματι
the on one hand to be in the kind or in the form one it is in the reality

ἐν ᾧ κινεῖται, τὸ δ' ἐχομένην ἐν τῷ χρόνῳ, τὸ δ' ἁπλῶς μίαν

in which it is moved, the and being held in the time, the and simply one

ἐν ἅπασι τούτοις· καὶ ἐν ᾧ γὰρ ἓν δεῖ εἶναι καὶ ἄτομον,

in all these things; and in which indeed one must be and indivisible,

οἷον τὸ εἶδος, καὶ τὸ ὅτε, οἷον τὸν χρόνον ἕνα καὶ μὴ διαλείπειν,

such as the form, and the when, such as the time one and not to interrupt,

καὶ τὸ κινούμενον ἓν εἶναι μὴ κατὰ συμβεβηκός, ὥσπερ τὸ

and the moving one to be not according to contingent, just as the

λευκὸν μελαίνεσθαι καὶ Κορίσκον βαδίζειν (ἓν δὲ Κορίσκος καὶ

white to become black and Koriskon to walk (one but Koriskos and

λευκόν, ἀλλὰ κατὰ συμβεβηκός), [228a] μηδὲ κοινόν· εἴη

white, but according to contingent), [228a] nor common; it might be

γὰρ ἂν ἅμα δύο ἀνθρώπους ὑγιάζεσθαι τὴν αὐτὴν ὑγίανσιν,

for indeed at once two people to be healthy the same health,

οἷον ὀφθαλμίας· ἀλλ' οὐ μία αὕτη, ἀλλ' εἴδει μία. τὸ δὲ Σωκράτη

such as of the eyes; but not one this, but in kind one. the and Socrates

τὴν αὐτὴν μὲν ἀλλοίωσιν ἀλλοιοῦσθαι τῷ εἴδει, ἐν ἄλλῳ

the same on the one hand alteration to be altered to the form, in another

δὲ χρόνῳ καὶ πάλιν ἐν ἄλλῳ, εἰ μὲν ἐνδέχεται τὸ φθαρὲν

and time and again in another, if on the one hand it is possible the perishable

πάλιν ἓν γίγνεσθαι τῷ ἀριθμῷ, εἴη ἂν καὶ αὕτη μία, εἰ

again one to become to the number, it would be perhaps and this one, if

δὲ μή, ἡ αὐτὴ μέν, μία δ' οὔ. ἔχει δ' ἀπορίαν

on the other hand not, the same on the one hand, one but no. has but a perplexity

ταύτῃ παραπλησίαν καὶ πότερον μία ἡ ὑγίεια καὶ ὅλως αἱ

in this similar and whether one the health and altogether the

ἕξεις καὶ τὰ πάθη τῇ οὐσίᾳ εἰσὶν ἐν τοῖς σώμασιν· κινούμενα

states and the passions to the essence are in the bodies. Moving

γὰρ φαίνεται τὰ ἔχοντα καὶ ῥέοντα. εἰ δὴ ἡ αὐτὴ καὶ μία ἡ

for it seems the possessing and flowing. if indeed the same and one the

ἔωθεν καὶ νῦν ὑγίεια, διὰ τί οὐκ ἂν καὶ ὅταν διαλιπὼν

from the dawn and now health, through why not would and when having departed

λάβῃ πάλιν τὴν ὑγίειαν, καὶ αὕτη κἀκείνη μία τῷ ἀριθμῷ

may receive again the health, and this and that one to the number

ἂν εἴη; ὁ γὰρ αὐτὸς λόγος· πλὴν τοσοῦτον διαφέρει, ὅτι

would be? the for same reason. but to such an extent it differs, that

εἰ μὲν δύο, δι' αὐτὸ τοῦτο, ὡς τῷ ἀριθμῷ, καὶ

if on the one hand two, through this same thing, as to the number, and

τὰς ἐνεργείας ἀνάγκη (μία γὰρ ἀριθμῷ ἐνέργεια ἑνὸς ἀριθμῷ)·

the activities necessity (one for number activity of one number).

εἰ δ' ἡ ἕξις μία, ἴσως οὐκ ἄν τῳ δόξειέ πω μία

if on the other hand the state is one, perhaps not would it seem at all one

καὶ ἡ ἐνέργεια εἶναι (ὅταν γὰρ παύσηται βαδίζων, οὐκέτι ἔστιν

and the activity to be (when for it ceases walking, no longer is

ἡ βάδισις, πάλιν δὲ βαδίζοντος ἔσται). εἰ δ' οὖν

the walking, again but of one walking there will be). if on the other hand then

μία καὶ ἡ αὐτή, ἐνδέχοιτ' ἂν τὸ αὐτὸ καὶ ἓν καὶ φθείρεσθαι καὶ

one and the same, it may be if the same and one and to be destroyed and

εἶναι πολλάκις. αὗται μὲν οὖν εἰσιν αἱ ἀπορίαι ἔξω τῆς νῦν

to be many times. these indeed therefore are the difficulties outside of the now

σκέψεως·

thought.

ἐπεὶ δὲ συνεχὴς πᾶσα κίνησις, τήν τε ἁπλῶς μίαν ἀνάγκη καὶ συνεχῆ

since but continuous every motion, the and simply one necessity and continuous

εἶναι, εἴπερ πᾶσα διαιρετή, καὶ εἰ συνεχής, μίαν. οὐ γὰρ πᾶσα
to be, if indeed every divisible, and if continuous, one. not for every
γένοιτ' ἂν συνεχὴς πάσῃ, ὥσπερ οὐδ' ἄλλο οὐδὲν
would come to be if continuous in every way, just as nor anything else nothing
τῷ τυχόντι τὸ τυχόν, ἀλλ' ὅσων ἓν τὰ ἔσχατα. ἔσχατα δὲ
by the happening the happening, but of those one the last things. last things but
τῶν μὲν οὐκ ἔστι, τῶν δ' ἔστιν ἄλλα τῷ εἴδει καὶ ὁμώνυμα·
of the indeed not is, of the but there are other by the kind and homonyms.
πῶς γὰρ ἂν ἅψαιτο ἢ ἓν γένοιτο τὸ ἔσχατον γραμμῆς καὶ
how for if it would touch or one would come to be the last of a line and
βαδίσεως; ἐχόμεναι μὲν οὖν εἶεν ἂν καὶ αἱ μὴ αἱ αὐταὶ
of movement? being held indeed therefore they would be if and the not the same
τῷ εἴδει μηδὲ τῷ γένει (δραμὼν γὰρ ἄν τις πυρέξειεν
by the kind nor by the genus having run for if anyone were to catch fire
εὐθύς), καὶ οἷον ἡ λαμπὰς <ἡ> ἐκ διαδοχῆς φορὰ ἐχομένη,
immediately and such as the lamp the from succession course being carried
συνεχὴς δ' οὔ. κεῖται γὰρ τὸ συνεχές, ὧν τὰ ἔσχατα ἕν. ὥστ'
continuous and not lies for the continuous of which the extremes are one so
ἐχόμεναι καὶ ἐφεξῆς εἰσὶ τῷ τὸν χρό νον εἶναι συνεχῆ, συνεχὴς
holding and subsequently are to the the time - to be continuous continuous
δὲ τῷ τὰς κινήσεις· τοῦτο δ', [228b] ὅταν ἓν τὸ ἔσχατον
but to the the movements this and [228b] whenever one the extreme
γένηται ἀμφοῖν. διὸ ἀνάγκη τὴν αὐτὴν εἶναι τῷ εἴδει καὶ ἑνὸς
happens to both therefore necessity the same to be to the kind and of one
καὶ ἐν ἑνὶ χρόνῳ τὴν ἁπλῶς συνεχῆ κίνησιν καὶ μίαν, τῷ χρόνῳ
and in one time the simply continuous motion and one, in the time
μέν, ὅπως μὴ ἀκινησία μεταξὺ ᾖ (ἐν τῷ διαλείποντι
on the one hand, so that not immobility between may be (in the interrupting

γὰρ ἡρεμεῖν ἀνάγκη· πολλαὶ οὖν καὶ οὐ μία ἡ κίνησις,

for tranquility necessity; many therefore and not one the movement,

ὧν ἐστὶν ἡρεμία μεταξύ, ὥστε εἴ τις κίνησις στάσει

of which there is tranquility among, so that if anyone movement is in a state

διαλαμβάνεται, οὐ μία οὐδὲ συνεχής· διαλαμβάνεται δέ, εἰ μεταξὺ

is perceived, not one nor continuous; it is perceived but, if between

χρόνος)· τῆς δὲ τῷ εἴδει μὴ μιᾶς, καὶ εἰ μὴ διαλείπεται [ὁ

time); of the and the form not one, and if not it is interrupted [the

χρόνος], ὁ μὲν [γὰρ] χρόνος εἷς, τῷ εἴδει δ' ἡ κίνησις ἄλλη·

time], the one [for] time is one, the form but the movement is another;

τὴν μὲν γὰρ μίαν ἀνάγκη καὶ τῷ εἴδει μίαν εἶναι, ταύτην δ' ἁπλῶς

this one for one necessity and the form one to be, this but simply

μίαν οὐκ ἀνάγκη. τίς μὲν οὖν κίνησις ἁπλῶς μία, εἴρηται·

one not a necessity. Who indeed therefore movement simply one, has been stated;

ἔτι δὲ λέγεται μία καὶ ἡ τέλειος, ἐάν τε κατὰ γένος ἐάν

moreover and it is said one and the perfect, if and according to kind if

τε κατ' εἶδος ᾖ ἐάν τε κατ' οὐσίαν, ὥσπερ καὶ ἐπὶ τῶν

and according to form it is if and according to substance, just as and upon the

ἄλλων τὸ τέλειον καὶ ὅλον τοῦ ἑνός. ἔστι δ' ὅτε κἂν ἀτελὴς

others the perfect and whole of the one. there is but when even imperfect

ᾖ μία λέγεται, ἐὰν μόνον ᾖ συνεχής. ἔτι δ' ἄλλως παρὰ

it is one is called, if only it is continuous. moreover but otherwise than

τὰς εἰρημένας λέγεται μία κίνησις ἡ ὁμαλής. ἡ γὰρ ἀνώμαλος

the stated is called one motion the smooth. the for irregular

ἔστιν ὡς οὐ δοκεῖ μία, ἀλλὰ μᾶλλον ἡ ὁμαλής, ὥσπερ ἡ εὐθεῖα·

is as not it seems one, but rather the smooth, just as the straight.

ἡ γὰρ ἀνώμαλος διαιρετή. ἔοικε δὲ διαφέρειν ὡς τὸ μᾶλλον καὶ

the for irregular is divisible. it seems but to differ as the more and

ἧττον. ἔστιν δὲ ἐν ἁπάσῃ κινήσει τὸ ὁμαλῶς ἢ μή· καὶ γὰρ ἂν
less. there is but in in every movement the smoothly or not and for if
ἀλλοιοῖτο ὁμαλῶς, καὶ φέροιτο ἐφ' ὁμαλοῦ οἷον κύκλου ἢ
it would change smoothly, and it would be carried upon a smooth like a circle or
εὐθείας, καὶ περὶ αὔξησιν ὡσαύτως καὶ φθίσιν. ἀνωμαλία δ'
a straight line, and regarding increase likewise and decrease. Irregularity but
ἐστὶν διαφορὰ ὁτὲ μὲν ἐφ' ᾧ κινεῖται (ἀδύνατον γὰρ ὁμαλὴν
is a difference whenever indeed upon which it moves (impossible for a smooth
εἶναι τὴν κίνησιν μὴ ἐπὶ ὁμαλῷ μεγέθει, οἷον ἡ τῆς κεκλασμένης
to be the motion not at a smooth magnitude, like the of the broken
κίνησις ἢ ἡ τῆς ἕλικος ἢ ἄλλου μεγέθους, ὧν μὴ ἐφαρμόττει
motion or the of the helix or of another magnitude, of which not applies
τὸ τυχὸν ἐπὶ τὸ τυχὸν μέρος)· ἡ δὲ οὔτε ἐν τῷ ὅ οὔτ' ἐν
the chance upon the chance part). The but neither in the which nor in
τῷ πότε οὔτε ἐν τῷ εἰς ὅ, ἀλλ' ἐν τῷ ὡς. ταχυτῆτι γὰρ
the when nor in the into that, but in the manner. quickness for
καὶ βραδυτῆτι ἐνίοτε διώρισται· ἧς μὲν γὰρ τὸ αὐτὸ
and slowness sometimes is distinguished; of which indeed for the same
τάχος, ὁμαλής, ἧς δὲ μή, ἀνώμαλος. διὸ οὐκ εἴδη κινήσεως
speed, smooth, of which but not, irregular. therefore not kinds of motion
οὐδὲ διαφοραὶ τάχος καὶ βραδυτής, ὅτι πάσαις ἀκολουθεῖ ταῖς
nor differences of speed and slowness, because all follows the
διαφόροις κατ' εἶδος. ὥστε οὐδὲ βαρύτης καὶ κουφότης ἡ εἰς
different by type. so that not even heaviness and lightness the into
τὸ αὐτό, οἷον γῆς πρὸς αὐτὴν ἢ πυρὸς πρὸς [229a] αὐτό. μία
the same, like of earth towards itself or of fire towards [229a] itself. one
μὲν οὖν ἡ ἀνώμαλος τῷ συνεχὴς <εἶναι>, ἧττον δέ, ὅπερ
indeed therefore the irregular by continuous <to be>, less however, which

τῇ κεκλασμένῃ συμβαίνει φορᾷ· τὸ δ' ἧττον μίξις ἀεὶ τοῦ
in the broken occurs by nature; the but less mixture always of the
ἐναντίου. εἰ δὲ πᾶσαν τὴν μίαν ἐνδέχεται καὶ ὁμαλὴν εἶναι καὶ μή,
opposite. if but all the one is possible and smooth to be and not,
οὐκ ἂν εἴησαν αἱ ἐχόμεναι αἱ μὴ κατ' εἶδος αἱ αὐταὶ μία
not would they be the possessing the not according to form the same one
καὶ συνεχής· πῶς γὰρ ἂν εἴη ὁμαλὴς ἡ ἐξ ἀλλοιώ σεως
and continuous. how for would it be smooth the from alteration of senses
συγκειμένη καὶ φορᾶς; δέοι γὰρ ἂν ἐφαρμόττειν.
composed and of forms? it is necessary for would to apply.

Κεφάλαιο 5

Chapter 5

Ἔτι δὲ διοριστέον ποία κίνησις ἐναντία κινήσει, καὶ περὶ
Yet but must be defined what kind of movement opposite movement, and concerning
μονῆς δὲ τὸν αὐτὸν τρόπον. διαιρετέον δὲ πρῶτον πότερον ἐναντία
rest but the same manner. must be divided but first whether opposite
κίνησις ἡ ἐκ τοῦ αὐτοῦ τῇ εἰς τὸ αὐτό (οἷον ἡ ἐξ ὑγιείας
movement the from the same to into the same (such as the from health
τῇ εἰς ὑγίειαν), οἷον καὶ γένεσις καὶ φθορὰ δοκεῖ, ἢ ἡ ἐξ
into into health), such as and generation and decay seems, or the from
ἐναντίων (οἷον ἡ ἐξ ὑγιείας τῇ ἐκ νόσου), ἢ ἡ εἰς ἐναντία
opposites (such as the from health into from disease), or the into opposite
(οἷον ἡ εἰς ὑγίειαν τῇ εἰς νόσον), ἢ ἡ ἐξ ἐναντίου τῇ εἰς
(such as the into health into into disease), or the from opposite to into
ἐναντίον (οἷον ἡ ἐξ ὑγιείας τῇ εἰς νόσον), ἢ ἡ ἐξ ἐναντίου
opposition (such as the from health into into disease), or the from opposition

εἰς ἐναντίον τῇ ἐξ ἐναντίου εἰς ἐναντίον (οἷον ἡ ἐξ ὑγιείας εἰς
into against the from opposition into against (such as the from health into
νόσον τῇ ἐκ νόσου εἰς ὑγίειαν). ἀνάγκη γὰρ ἢ ἕνα τινὰ τούτων
disease the from disease into health). necessity for or one certain of these
εἶναι τῶν τρόπων ἢ πλείους· οὐ γὰρ ἔστιν ἄλλως ἀντιτιθέναι.
to be the ways or more; not for is otherwise to oppose.

ἔστι δ' ἡ μὲν ἐξ ἐναντίου τῇ εἰς ἐναντίον οὐκ ἐναντία, οἷον ἡ
is but the one from the opposite the into opposite not opposite, such as the
ἐξ ὑγιείας τῇ εἰς νόσον· ἡ αὐτὴ γὰρ καὶ μία. τὸ μέντοι γ' εἶναι
from health the into disease. the same for and one. the however it is
οὐ ταὐτὸ αὐταῖς, ὥσπερ οὐ ταὐτὸ τὸ ἐξ ὑγιείας μεταβάλλειν καὶ
not the same with them, just as not the same the from health changing and
τὸ εἰς νόσον. οὐδ' ἡ ἐξ ἐναντίου τῇ ἐξ ἐναντίου· ἅμα
the into disease. nor the from the opposite the from the opposite; at the same time
μὲν γὰρ συμβαίνει ἐξ ἐναντίου καὶ εἰς ἐναντίον ἢ μεταξύ—ἀλλὰ
on the one hand for it happens from the opposite and into the opposite or between-but
περὶ τούτου μὲν ὕστερον ἐροῦμεν, ἀλλὰ μᾶλλον τὸ εἰς
about this on one hand later we will say, but rather the into
ἐναντίον μεταβάλλειν δόξειεν ἂν εἶναι αἴτιον τῆς ἐναντιώσεως
opposite changing would seem would to be the cause of the opposition
ἢ τὸ ἐξ ἐναντίου· ἡ μὲν γὰρ ἀπαλλαγὴ ἐναντιότητος,
or the from the opposite; the on the one hand for release from opposition,
ἡ δὲ λῆψις. καὶ λέγεται δ' ἑκάστη εἰς ὃ μεταβάλ λει μᾶλλον
the but grasp. and it is said that each into what change is rather
ἢ ἐξ οὗ, οἷον ὑγίανσις ἡ εἰς ὑγίειαν, νόσανσις δ' ἡ
than out of which, such as health the into health, disease that the

εἰς νόσον. λείπεται δὴ ἡ εἰς ἐναντία καὶ ἡ εἰς ἐναντία ἐξ
into disease. it remains indeed the into opposite and the into opposite out of
ἐναντίων. τάχα μὲν οὖν συμβαίνει τὰς εἰς ἐναντία καὶ
opposites. perhaps on the one hand therefore it happens the into opposite and
ἐξ ἐναντίων εἶναι, ἀλλὰ τὸ εἶναι ἴσως οὐ ταὐτό, λέγω δὲ τὸ εἰς
out of opposites to be, but the being perhaps not the same, I say but the into
ὑγίειαν τῷ ἐκ νόσου καὶ τὸ ἐξ ὑγιείας τῷ εἰς νόσον.
health to the from disease and the out of health to the into disease.

ἐπεὶ δὲ διαφέρει μεταβολὴ κινήσεως (ἡ ἔκ τινος γὰρ ὑποκειμένου
since but differs change of movement the out of something for underlying
εἴς τι ὑποκείμενον μεταβολὴ κίνησίς ἐστιν), ἡ ἐξ ἐναντίου
into something underlying change of movement is), the from opposite
[229b] εἰς ἐναντίον τῇ ἐξ ἐναντίου εἰς ἐναντίον κίνησις ἐναντία,
[229b] into opposition the from opposite into opposition motion opposite,
οἷον ἡ ἐξ ὑγιείας εἰς νόσον τῇ ἐκ νόσου εἰς ὑγίειαν. δῆλον
such as the from health into disease the out of disease into health. clear
δὲ καὶ ἐκ τῆς ἐπαγωγῆς ὁποῖα δοκεῖ τὰ ἐναντία εἶναι· τὸ
but also from the induction of what kind seems the opposites to be; the
νοσάζεσθαι γὰρ τῷ ὑγιάζεσθαι καὶ τὸ μανθάνειν τῷ ἀπατᾶσθαι
being ill for by being healthy and the learning by being deceived
μὴ δι' αὐτοῦ (εἰς ἐναντία γάρ· ὥσπερ γὰρ ἐπιστήμην, ἔστι
not through itself (into opposition for; just as for knowledge, there is
καὶ ἀπάτην καὶ δι' αὐτοῦ κτᾶσθαι καὶ δι' ἄλλου), καὶ ἡ ἄνω
and deceit and through itself to possess and through another), and the upper
φορὰ τῇ κάτω (ἐναντία γὰρ ταῦτα ἐν μήκει), καὶ ἡ εἰς δεξιὰ τῇ εἰς
part in lower (opposite for these in length), and the to right in to

ἀριστερά (ἐναντία γὰρ ταῦτα ἐν πλάτει), καὶ ἡ εἰς τὸ ἔμπροσθεν

left (opposite for these in width), and the to the front

τῇ εἰς τὸ ὄπισθεν (ἐναντία γὰρ καὶ ταῦτα). ἡ δ' εἰς ἐναντίον

in to the back (opposite for and these). the but to opposite

μόνον οὐ κίνησις ἀλλὰ μεταβολή, οἷον τὸ γίγνεσθαι λευκὸν

only not motion but change, such as the becoming white

μὴ ἔκ τινος. καὶ ὅσοις δὲ μὴ ἔστιν ἐναντία, ἡ ἐξ αὐτοῦ τῇ

not from anything. and to those but not is opposite, the from itself in

εἰς αὐτὸ μεταβολῇ ἐναντία· διὸ γένεσις φθορᾷ ἐναντία καὶ

to it change opposite· therefore generation decay opposite and

ἀποβολὴ λήψει· αὗται δὲ μεταβολαὶ μέν, κινήσεις δ' οὔ. τὰς δ'

loss will take· these but changes indeed, motions but not. the that

εἰς τὸ μεταξὺ κινήσεις, ὅσοις τῶν ἐναντίων ἔστι μεταξύ, ὡς

into the between movements, to those of the opposites there is among, as

εἰς ἐναντία πως θετέον· ὡς ἐναντίῳ γὰρ χρῆται τῷ

into opposition somehow it must be set; as to the opposite for it is used with the

μεταξὺ ἡ κίνησις, ἐφ' ὁπότερα ἂν μεταβάλλῃ, οἷον ἐκ φαιοῦ

between the movement, whenever whichever it may change, such as from brown

μὲν εἰς τὸ λευκὸν ὡς ἐκ μέλανος, καὶ ἐκ λευκοῦ εἰς

on the one hand into the white as from black, and from white into

φαιὸν ὡς εἰς μέλαν, ἐκ δὲ μέλανος εἰς φαιὸν ὡς εἰς λευκὸν τὸ

gray as into black, from but black into gray as into white the

φαιόν· τὸ γὰρ μέσον πρὸς ἑκάτερον λέγεταί πως ἑκάτερον τῶν

gray; the for middle towards each is said somehow each of the

ἄκρων, καθάπερ εἴρηται καὶ πρότερον. κίνησις μὲν δὴ

extremes, just as it has been said and before. movement on the one hand indeed

κινήσει ἐναντία οὕτως ἡ ἐξ ἐναντίου εἰς ἐναντίον τῇ ἐξ

will move oppositely in this way the from the opposite into opposition to the out of

ἐναντίου εἰς ἐναντίον.

opposite into against.

Κεφάλαιο 6

Chapter 6

Ἐπεὶ δὲ κινήσει οὐ μόνον δοκεῖ κίνησις εἶναι ἐναντία ἀλλὰ καὶ

Since but movement not only seems motion to be contrary but and

ἠρεμία, τοῦτο διοριστέον. ἁπλῶς μὲν γὰρ ἐναντίον κίνησις

tranquility, this must be defined. Simply on one hand for opposite motion

κινήσει, ἀντίκειται δὲ καὶ ἠρεμία (στέρησις γάρ, ἔστι δ'

moves, stands in opposition but and tranquility (deprivation for, it is that

ὡς καὶ ἡ στέρησις ἐναντία λέγεται), ποιᾷ δὲ ποιά, οἷον

as also the deprivation opposite is said), in what but what kind, such as

τῇ κατὰ τόπον ἡ κατὰ τόπον. ἀλλὰ τοῦτο νῦν λέγεται

the according to place the according to place. But this now is said

ἁπλῶς· πότερον γὰρ τῇ ἐνταῦθα μονῇ ἡ ἐκ τούτου ἢ ἡ εἰς

simply: whether for the here alone the from this or the into

τοῦτο κίνησις ἀντίκειται; δῆλον δὴ ὅτι, ἐπεὶ ἐν δυσὶν ἡ

this motion is opposed? It is clear indeed that, since in two the

κίνησις ὑποκειμένοις, τῇ μὲν ἐκ τούτου εἰς τὸ ἐναντίον ἡ ἐν

motion in subjects, the on one hand from this to the opposite the in

τούτῳ μονή, τῇ δ' ἐκ τοῦ ἐναντίου εἰς τοῦτο ἡ ἐν τῷ ἐναντίῳ.

this dwelling, the but from the opposite to this the in the opposite.

ἅμα δὲ καὶ ἀλλήλαις ἐναντίαι αὗται· καὶ γὰρ ἄτοπον,

at the same time but also to one another opposite these; and for it is absurd,

εἰ κινήσεις [230a] μὲν ἐναντίαι εἰσίν, ἠρεμίαι δ' ἀντικείμεναι οὐκ

if movements [230a] indeed opposite they are, calmness but opposing not

εἰσίν. εἰσὶν δὲ αἱ ἐν τοῖς ἐναντίοις, οἷον ἡ ἐν ὑγιείᾳ τῇ ἐν

they are. they are but the in the opposites, such as the in health the in

νόσῳ ἠρεμίᾳ (κινήσει δὲ τῇ ἐξ ὑγιείας εἰς νόσον· τῇ γὰρ ἐκ

disease calmness (movement but the from health to disease; the for from

νόσου εἰς ὑγίειαν ἄλογον—ἡ γὰρ εἰς αὐτὸ κίνησις ἐν ᾧ

disease to health it is irrational-the for to itself movement in which

ἔστηκεν, ἠρέμησις μᾶλλόν ἐστιν, ἢ συμβαίνει γε ἅμα

it has stood, calmness rather it is, than it happens indeed at the same time

γίγνεσθαι τῇ κινήσει ἀνάγκη δὲ ἢ ταύτην ἢ ἐκείνην εἶναι)· οὐ γὰρ

to happen the movement necessity but or this or that to be) not for

ἥ γ' ἐν λευκότητι ἠρεμία ἐναντία τῇ ἐν ὑγιείᾳ.

which is in whiteness serenity opposite to the in health.

ὅσοις δὲ μὴ ἔστιν ἐναντία, τούτων μεταβολὴ μέν ἐστιν ἀντικειμένη

to those but not it is contrary, of these change on one hand it is opposed

ἡ ἐξ αὐτοῦ τῇ εἰς αὐτό, κίνησις δ' οὐκ ἔστιν, οἷον ἡ

the from it to the into itself, motion but not it is, such as the

ἐξ ὄντος τῇ εἰς ὄν, καὶ μονὴ μὲν τούτων οὐκ ἔστιν,

from being to the into being, and rest on one hand of these not it is,

ἀμεταβλησία δέ. καὶ εἰ μέν τι εἴη ὑποκείμενον,

unchangeability on the other hand. and if on one hand something were a subject,

ἡ ἐν τῷ ὄντι ἀμεταβλησία τῇ ἐν τῷ μὴ ὄντι ἐναντία. εἰ δὲ μὴ

the in the being unchangeability to the in the not being contrary. if but not

ἔστι τι τὸ μὴ ὄν, ἀπορήσειεν ἄν τις τίνι ἐναντία ἡ

it is something the not being, it would perplex someone certain to whom contrary the

ἐν τῷ ὄντι ἀμεταβλησία, καὶ εἰ ἠρεμία ἐστίν. εἰ δὲ τοῦτο, ἢ οὐ

in the being unchangeability, and if tranquility exists. if but this, or not

πᾶσα ἠρεμία κινήσει ἐναντία, ἢ ἡ γένεσις καὶ φθορὰ κίνησις.

all tranquility motion opposing, or the generation and decay motion.

δῆλον τοίνυν ὅτι ἠρεμία μὲν οὐ λεκτέα, εἰ μὴ καὶ αὗται

evident therefore that tranquility on one hand not to be spoken of, if not and these

κινήσεις, ὅμοιον δέ τι καὶ ἀμεταβλησία· ἐναντία δὲ ἢ οὐδενὶ

motions, similar but something and immutability; opposing but or to no one

ἢ τῇ ἐν τῷ μὴ ὄντι ἢ τῇ φθορᾷ· αὕτη γὰρ ἐξ αὐτῆς, ἡ δὲ

or to the in the not being or to the decay; this for out of it, the and

γένεσις εἰς ἐκείνην.

generation into that one.

ἀπορήσειε δ' ἄν τις διὰ τί ἐν μὲν τῇ κατὰ τόπον μεταβολῇ

will wonder but if anyone for what in indeed the according to place change

εἰσὶ καὶ κατὰ φύσιν καὶ παρὰ φύσιν καὶ μοναὶ καὶ κινήσεις,

are and according to nature and contrary to nature and singular and motions,

ἐν δὲ ταῖς ἄλλαις οὔ, οἷον ἀλλοίωσις ἡ μὲν κατὰ φύσιν ἡ

in but the other not, such as alteration the indeed according to nature the

δὲ παρὰ φύσιν (οὐδὲν γὰρ μᾶλλον ἡ ὑγίανσις ἢ ἡ νόσανσις

but contrary to nature nothing for more the healing or the disease

κατὰ φύσιν ἢ παρὰ φύσιν, οὐδὲ λεύκανσις ἢ μέλανσις)·

according to nature or contrary to nature, nor whitening or blackening)·

ὁμοίως δὲ καὶ ἐπ' αὐξήσεως καὶ φθίσεως (οὔτε γὰρ αὗται

similarly but also in relation to increase and decrease neither for these

ἀλλήλαις ἐναντίαι ὡς φύσει ἡ δὲ παρὰ φύσιν, οὔτ' αὔξησις

are contrary opposites as by nature the but contrary to nature, nor increase

αὐξή σει)· καὶ ἐπὶ γενέσεως δὲ καὶ φθορᾶς ὁ αὐτὸς λόγος·

grows is)· and upon generation but and of decay the same word·

οὔτε γὰρ ἡ μὲν γένεσις κατὰ φύσιν ἡ δὲ φθορὰ
neither for the on the one hand generation according to nature the but decay
παρὰ φύσιν (τὸ γὰρ γηρᾶν κατὰ φύσιν), οὔτε γένεσιν ὁρῶμεν
against nature (that for aging according to nature), neither generation do we see
τὴν μὲν κατὰ φύσιν τὴν δὲ παρὰ φύσιν. ἢ εἰ ἔστιν τὸ
the on the one hand according to nature the but against nature. Or if there is the
βίᾳ παρὰ φύσιν, καὶ φθορὰ ἂν εἴη φθορᾷ ἐναντία ἡ βίαιος ὡς
by force against nature, and decay would be of decay opposite the violent as
παρὰ φύσιν οὖσα τῇ κατὰ φύσιν; ἆρ' οὖν καὶ γενέσεις εἰσὶν
against nature being the according to nature? Is it then and generations are
ἔνιαι βίαιοι καὶ οὐχ εἱμαρμέναι, αἷς ἐναντίαι αἱ κατὰ φύσιν,
some violent and not fated, which are opposite the according to nature,
[230b] καὶ αὐξήσεις βίαιοι καὶ φθίσεις, οἷον αὐξήσεις αἱ τῶν ταχὺ
[230b] and increases violent and declines, such as increases the of the quickly
διὰ τρυφὴν ἡβώντων, καὶ οἱ σῖτοι οἱ ταχὺ ἀδρυνόμενοι καὶ
through indulgence of the youthful, and the foods the quickly being made rich and
μὴ πιληθέντες; ἐπὶ δ' ἀλλοιώσεως πῶς; ἢ ὡσαύτως; εἶεν
not having been made fat? upon but of alteration how? or likewise? they might be
γὰρ ἄν τινες βίαιοι, αἱ δὲ φυσικαί, οἷον οἱ ἀφιέμενοι μὴ
for if some violent, the and the natural, such as the those letting go not
ἐν κρισίμοις ἡμέραις, οἱ δ' ἐν κρισίμοις· οἱ μὲν οὖν
in critical days, the but in critical ones. the on one hand therefore
παρὰ φύσιν ἠλλοίωνται, οἱ δὲ κατὰ φύσιν. ἔσονται δὴ καὶ
alongside nature are altered, the but according to nature. they will be indeed and
φθοραὶ ἐναντίαι ἀλλήλαις, οὐ γενέσεσι. καὶ τί γε κωλύει
destructions contrary to each other, not in generations. and what indeed prevents
ἔστιν ὡς; καὶ γὰρ εἰ ἡ μὲν ἡδεῖα ἡ δὲ λυπηρὰ εἴη· ὥστε
it is as? and for if the on one hand pleasant the but painful would be; so that

οὐχ ἁπλῶς φθορὰ φθορᾷ ἐναντία, ἀλλ' ᾗ ἡ μὲν
not simply destruction to destruction contrary to, but where the on the one hand
τοιαδὶ ἡ δὲ τοιαδὶ αὐτῶν ἐστιν.
such the but such of them it is.

ὅλως μὲν οὖν ἐναντίαι κινήσεις καὶ ἠρεμίαι τὸν εἰρημένον
altogether on one hand therefore opposing movements and rest the mentioned
τρόπον εἰσίν, οἷον ἡ ἄνω τῇ κάτω· τόπου γὰρ ἐναντιώσεις
manner are, such as the upper the lower. of places for oppositions
αὗται. φέρεται δὲ τὴν μὲν ἄνω φορὰν φύσει τὸ πῦρ, τὴν
these. it is said but the on one hand upper motion by nature the fire, the
δὲ κάτω ἡ γῆ· καὶ ἐναντίαι γ' αὐτῶν αἱ φοραί.
on the other hand lower the earth. and opposing indeed of them the movements.
τὸ δὲ πῦρ ἄνω μὲν φύσει, κάτω δὲ παρὰ φύσιν·
the on the other hand fire upward indeed by nature, downward but contrary to nature.
καὶ ἐναντία γε ἡ κατὰ φύσιν αὐτοῦ τῇ παρὰ φύσιν.
and opposing indeed the according to nature of it the contrary to nature.
καὶ μοναὶ δ' ὡσαύτως· ἡ γὰρ ἄνω μονὴ τῇ ἄνωθεν κάτω
and singular but likewise; the for upper dwelling the from above downward
κινήσει ἐναντία. γίγνεται δὲ τῇ γῇ ἡ μὲν μονὴ ἐκείνη
motion opposing. it happens but the earth the on one hand dwelling that
παρὰ φύσιν, ἡ δὲ κίνησις αὕτη κατὰ φύσιν. ὥστε κινήσει
contrary to nature, the but movement this according to nature. so that in motion
μονὴ ἐναντία ἡ παρὰ φύσιν τῇ κατὰ φύσιν τοῦ αὐτοῦ·
a single opposite the contrary to nature to the according to nature of the same;
καὶ γὰρ ἡ κίνησις ἡ τοῦ αὐτοῦ ἐναντία οὕτως· ἡ μὲν
and indeed the movement the of the same opposite in this way; the on one hand

γὰρ κατὰ φύσιν [ἔσται] αὐτῶν, ἡ ἄνω ἢ ἡ κάτω, ἡ δὲ
indeed according to nature [will be] of them, the upward or the downward, the but
παρὰ φύσιν. ἔχει δ' ἀπορίαν εἰ ἔστιν πάσης ἠρεμίας τῆς μὴ
contrary to nature. has but a lack if it is of all rest, of the not
ἀιεὶ γένεσις, καὶ αὕτη τὸ ἵστασθαι. τοῦ δὴ παρὰ φύσιν
always generation, and this the standing still. of the indeed contrary to nature
μένοντος, οἷον τῆς γῆς ἄνω, εἴη ἂν γένεσις. ὅτε ἄρα
remaining, such as of the earth upward, would be if generation. when therefore
ἐφέρετο ἄνω βίᾳ, ἵστατο. ἀλλὰ τὸ ἱστάμενον ἀεὶ δοκεῖ
it was brought upward by force, it stood still. but the standing always seems
φέρεσθαι θᾶττον, τὸ δὲ βίᾳ τοὐναντίον. οὐ γενόμενον ἄρα
to be carried faster, the but by force contrary. not having come to be therefore
ἠρεμοῦν ἔσται ἠρεμοῦν. ἔτι δοκεῖ τὸ ἵστασθαι ἢ ὅλως εἶναι τὸ
will be at rest will be at rest. still seems the standing or entirely to be the
εἰς τὸν αὑτοῦ τόπον φέρεσθαι ἢ συμβαίνειν ἅμα.
into the one's own place to be carried or to happen together.

ἔχει δ' ἀπορίαν εἰ ἐναντία ἡ μονὴ ἡ ἐνταῦθα τῇ ἐντεῦθεν
has but a perplexity if opposite the motion the here to yonder
κινήσει· ὅταν γὰρ κινῆται ἐκ τουδὶ καὶ ἀποβάλλῃ, ἔτι δοκεῖ
moves. when for it moves from this and it casts off, still it seems
ἔχειν τὸ ἀποβαλλόμενον, ὥστ' εἰ αὕτη ἡ ἠρεμία ἐναντία τῇ
to have the thing being cast off, so that if this the rest is opposite to the
ἐντεῦθεν εἰς τοὐναντίον κινήσει, ἅμα ὑπάρξει τἀναντία.
yonder to the opposite motion, at the same time it will exist the opposites.
ἢ πῇ ἠρεμεῖ, εἰ ἔτι μένει, ὅλως δὲ τοῦ κινουμένου
or where it is at rest, if still it remains, altogether but of the moving

τὸ μὲν [231a] ἐκεῖ, τὸ δ' εἰς ὃ μεταβάλλει;

the on the one hand [231a] there, the on the other hand towards that it changes?

διὸ καὶ μᾶλλον κίνησις κινήσει ἐναντίον ἢ ἠρέμησις.

therefore also more motion will move opposite or rest.

καὶ περὶ μὲν κινήσεως καὶ ἠρεμίας, πῶς ἑκατέρα μία, καὶ

and regarding on the one hand of motion and of rest, how each one, and

τίνες ἐναντίαι τίσιν, εἴρηται.

what opposites by whom, is spoken of.

[ἀπορήσειε δ' ἄν τις καὶ περὶ τοῦ ἵστασθαι, εἰ καὶ ὅσαι

will wonder but if anyone and concerning the to stand, if and however many

παρὰ φύσιν κινήσεις, ταύταις ἔστιν ἠρεμία ἀντικειμένη. εἰ

contrary to nature motions, these there is tranquility opposing. if

μὲν οὖν μὴ ἔσται, ἄτοπον· μένει γάρ, βίᾳ δέ. ὥστε

indeed then not will be, absurd; it remains for by force however. so that

ἠρεμοῦν τι ἔσται οὐκ ἀεὶ ἄνευ τοῦ γενέσθαι. ἀλλὰ δῆλον

to be at rest something will be not always without the becoming. but clear

ὅτι ἔσται· ὥσπερ γὰρ κινεῖται παρὰ φύσιν, καὶ ἠρεμοίη

that it will be; just as for it moves contrary to nature, and it would be at rest

ἄν τι παρὰ φύσιν. ἐπεὶ δ' ἔστιν ἐνίοις κίνησις κατὰ

if something contrary to nature. since but there is in some motion according to

φύσιν καὶ παρὰ φύσιν, οἷον πυρὶ ἡ ἄνω κατὰ φύσιν ἡ δὲ

nature and contrary to nature, such as fire the upward according to nature the but

κάτω παρὰ φύσιν, πότερον αὕτη ἐναντία ἢ ἡ τῆς γῆς; αὕτη

downward contrary to nature, whether this is opposite or the of the earth? this

γὰρ φέρεται κατὰ φύσιν κάτω. ἢ δῆλον ὅτι ἄμφω, ἀλλ' οὐχ

for it is borne according to nature below. or it is clear that both, but not

ὡσαύτως, ἀλλ᾽ ἡ μὲν κατὰ φύσιν ὡς κατὰ φύσιν οὔσης

likewise, but the one according to nature as according to nature existing

τῆς αὐτοῦ· ἡ δ᾽ ἄνω τοῦ πυρὸς τῇ κάτω, ὡς ἡ κατὰ

of the same. the other above of the fire in the below, as the according to

φύσιν οὖσα τῇ παρὰ φύσιν οὔσῃ. ὁμοίως δὲ καὶ ταῖς μοναῖς.

nature being in the contrary to nature existing. similarly and also to the units.

ἴσως δ᾽ ἠρεμίᾳ κίνησίς πῃ ἀντίκειται.]

perhaps but in stillness motion somewhere opposes.